

Related Work Survey

Prior research relevant to Moodscape: geographic sonification, soundscape–emotion perception, Korean urban soundscape studies, place identity/cultural branding, and accessible sonification

This survey was compiled to answer a direct question Moodscape is likely to face when presented: has anyone published research directly matching this project's approach — converting a place's geographic and demographic data into music? The short answer, based on the search conducted for this document, is **no single paper does exactly that**. But the project sits at the intersection of several active, well-established research threads, listed below. Each entry is a real, independently verifiable publication with its citation; none of the source text is reproduced beyond short, clearly-marked summaries of published findings.

1. Data Sonification of Geographic & Emotional Data

The closest precedent to Moodscape's core idea — mapping a place's character to sound at city scale — is **Chatty Maps** (Aiello, Schifanella, Quercia & Aletta, 2016), which built sound maps of London and Barcelona from geotagged social-media photo tags, then linked those sound categories to emotional responses (joy, sadness, surprise) and to street-level perceptions such as chaotic, monotonous, calm, and exciting. It is the strongest existing example of sound + geography + emotion combined at an urban scale, though it sources its sound categories from photo tags rather than generating original music, and does not use demographic statistics (density, safety, climate) the way Moodscape does.

More broadly, data sonification — turning any dataset into non-speech audio — is an active field with its own dedicated tooling, e.g. the **STRAUSS** framework (Sonification Tools and Resources for Analysis Using Sound Synthesis, 2025) and NASA's own public sonification program (*From Data to Melody*, NASA Earthdata, 2026), which turns astronomical datasets into audio for both accessibility and public engagement.

2. Soundscape Perception Standards

Moodscape's own "mood score" (peaceful ↔ exciting) is not an invented axis — it closely parallels the **Soundscape Circumplex Model (SCM)** formalized in the international standard **ISO/TS 12913-2:2018**, itself an adaptation of Russell's (1980) psychological circumplex model of affect for the specific domain of environmental sound. The ISO model places eight perceptual descriptors — pleasant, vibrant, eventful, chaotic, annoying, monotonous, uneventful, calm — around two axes, *Pleasantness* and *Eventfulness*, each 45° apart on a circle. This is the standard instrument the soundscape research community actually uses to quantify how a place "feels" acoustically, and is the framework adopted for the emotional data in the companion report to this survey.

3. Korean Urban Soundscape Research

Several recent, directly relevant studies exist on Korean cities specifically:

- Hong, J.Y. & Jeon, J.Y. (2020). Comparing associations among sound sources, human behaviors, and soundscapes on central business and commercial streets in Seoul, Korea. *Building and Environment*, 186, 107327. Found that soundscapes dominated by human activity sounds correlated with positive emotional responses, while soundscapes on historically/culturally significant streets carried a higher share of "meaning-related" responses than purely commercial streets.
- Kim, G. et al. & Hong, J. (2025). Exploring associations between urban soundscape and contextual factors based on a textual data-driven analysis and mapping approach: A case study in Daejeon, Korea. *Applied Acoustics*, 228, 110319. Proposes a text-data-driven method (TD-CAMM) for mapping soundscape perception against contextual/demographic factors in Yuseong-gu, Daejeon — one of the 17 regions covered by Moodscape.
- A 2025/2026 PLOS One study ("Mapping urban soundscape patterns using user-generated content: A place model approach to acoustic environment perception") applied Canter's place model (physical setting + activity + meaning) to Korean-language social-media text from 156 points of interest across central and southeastern Seoul, finding historical districts carry a notably higher share of meaning-related soundscape mentions (22.0%) than modern commercial districts (13.4%) — direct quantitative support for treating a neighborhood's history/culture, not just its physical sound level, as part of its soundscape identity.

4. Place Identity, Cultural Character & City Branding

For the cultural/character half of Moodscape's data (as opposed to the numeric/emotional half), the relevant literature is **place identity** and **city branding** research rather than acoustics. Key concepts used to structure this survey's approach include *cultural imageability* — a city's capacity to conjure strong mental images through its cultural features, symbols, and stories (Lynch's imageability concept, extended to cultural content in recent systematic reviews) — and the observation, repeated across this literature, that place identity is not a fixed fact but something "negotiated through multi-stakeholder dialogues, narrative construction, and symbolic representation" (per a 2025 *Frontiers in Communication* study on cultural storytelling in a small mountain city). This supports treating the cultural/character write-up for each Moodscape region as a synthesized narrative grounded in real, sourced facts, rather than as raw statistics.

5. Accessible Sonification (Blind, Deaf & Hard-of-Hearing Users)

For the accessibility section of the companion report, the relevant baseline work includes **iSonic** (Zhao et al., ACM TOCHI, 2008), the closest academic precedent for a non-visual geographic data interface, which showed blind participants could successfully identify facts and trends in unfamiliar geo-referenced data through coordinated sound and speech output. More recent work extends this: a 2022–2025 survey of AI-powered sound-awareness systems for deaf and hard-of-hearing users catalogs wearable/mobile tools (SoundWatch, ProtoSound, SoundSafe, and others) that convert ambient sound into visual or haptic alerts — the inverse problem of Moodscape's own sound generation, and a reminder that any accessibility layer needs a non-auditory channel (visual captions, haptic feedback) for deaf/HoH users,

not just louder or clearer audio. Separately, a 2024 arXiv study (Exploring the Diversity of Music Experiences for Deaf and Hard of Hearing People) documents that many deaf and HoH listeners do experience music meaningfully through vibration, rhythm, and visual cues rather than through hearing pitch directly — relevant to how Moodscape's music could be paired with a synchronized visual/haptic layer rather than assumed to be audio-only.

6. Where Moodscape Sits Relative to This Work

No paper found in this search combines all of the following the way Moodscape does: (a) real geographic/demographic statistics (population density, safety index, climate, language proficiency) as the direct numeric input, (b) an established perceptual framework (the ISO 12913 circumplex) for the emotional axis, (c) a place's cultural/historical character rendered in natural language rather than only in numbers, and (d) actual generative music output rather than a sound map built from pre-recorded or tagged audio. Chatty Maps is the closest precedent for (a)+(a subset of the emotional axis), but does not generate music or use demographic data. iSonic is the closest precedent for treating this as an accessibility tool. Moodscape's specific fusion of these four elements — and its accessibility-first design intent — appears to be a genuine, if narrow, gap in the current literature, which is exactly the kind of finding worth stating plainly in a KSEF-style report rather than overselling as "the first-ever" anything.

Search methodology note: this survey was compiled via web search across Google Scholar-indexed venues, ScienceDirect, ACM Digital Library, arXiv, PLOS One, and MDPI in August 2026. It is a scoping survey, not a systematic review — it is unlikely to be fully exhaustive, and papers behind institutional paywalls that do not appear in general web search were not accessible for full-text verification (citations for those were cross-checked against at least two independent listing sources).

References

- Aiello, L. M., Schifanella, R., Quercia, D., & Aletta, F. (2016). Chatty maps: constructing sound maps of urban areas from social media data. *Royal Society Open Science*, 3(3), 150690. <https://doi.org/10.1098/rsos.150690>
- International Organization for Standardization. (2018). *ISO/TS 12913-2:2018 — Acoustics — Soundscape — Part 2: Data collection and reporting requirements*.
- Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161–1178.
- Hong, J. Y., & Jeon, J. Y. (2020). Comparing associations among sound sources, human behaviors, and soundscapes on central business and commercial streets in Seoul, Korea. *Building and Environment*, 186, 107327.
- Kim, G. et al., & Hong, J. (2025). Exploring associations between urban soundscape and contextual factors based on a textual data-driven analysis and mapping approach: A case study in Daejeon, Korea. *Applied Acoustics*, 228, 110319.
- [Authors not independently confirmed beyond the journal listing] (2025/2026). Mapping urban soundscape patterns using user-generated content: A place model approach to acoustic environment perception. *PLOS One*.
- Zhao, H., Plaisant, C., Shneiderman, B., & Lazar, J. (2008). Data sonification for users with visual impairment: A case study with georeferenced data (iSonic). *ACM Transactions on Computer-Human Interaction (TOCHI)*, 15(1), Article 4.
- STRAUSS: Sonification Tools and Resources for Analysis Using Sound Synthesis. (2025). arXiv:2504.01660.
- NASA Earthdata. (2026). From Data to Melody: Data Sonification and Its Role in Open Science. earthdata.nasa.gov
- [Author survey, 2025]. AI-Powered Sound Awareness Systems for Deaf and Hard-of-Hearing Individuals: A Survey of Recent Advances (2022–2025). ResearchGate preprint.
- Exploring the Diversity of Music Experiences for Deaf and Hard of Hearing People. (2024). arXiv:2401.09025.
- Cultural Imageability on Urban Identity: A Systematic Literature Review. (2024/2025). *International Journal of Research and Innovation in Social Science*.
- Shaping place identity through cultural storytelling: strategic communication in a small mountain city. (2025). *Frontiers in Communication*.